Consequences of Thermal Energy Transfer

Exam code: 0625 & 0972

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Updated on 24 September 2024

Simple consequences of energy transfer

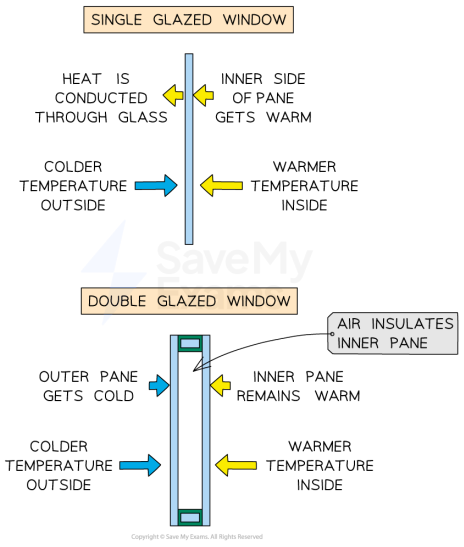
- Conduction, convection and radiation have many everyday applications and consequences

Examples of conduction

- Good conductors help transfer thermal energy **quickly**
- Examples include:
  - Metal pans to heat food quickly
  - Metal radiators to transfer heat from water inside to the surrounding air quickly

- Bad conductors (insulators) help retain thermal energy as they transfer heat **slowly**
- Examples include:
  - Plastic handles of saucepans to slow thermal energy transferred to hands
  - Air spaces in the walls or windows of some houses help to retain heat, as air is a poor conductor

Double-glazed windows



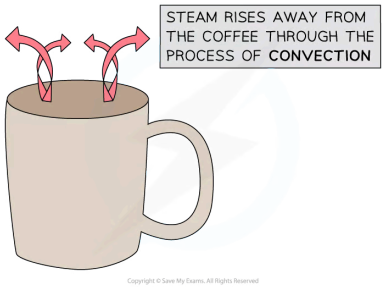
Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

Examples of convection

- Common applications of convection are:
  - heating a room with a radiator
  - steam rising which cools a hot liquid
- Radiators use convection to raise the temperature of a room in a building:
  1. The metal radiator is hot and transfers thermal energy to air nearby
  2. The particles of this hot air spread out, making it less dense
  3. The spread-out air is less dense than the air above it, so this hot air rises
  4. The radiator heats the cold air which replaces the hot air
  5. The newly heated air also rises, cools and sinks (as it contracts and increases in density)

Convection in steam from coffee



Thermal energy is transferred from the hot coffee to the air by convection currents rising from the surface

Examiner Tips and Tricks

Don't let the name 'radiator' confuse you, these should really be called 'convection heaters'. Their heating ability comes almost exclusively through convection, radiation plays a very small part.

Complex consequences of energy transfer

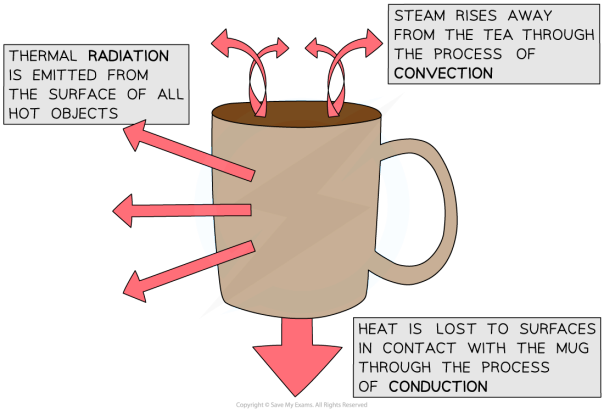
Extended tier only

Multiple paths of energy transfer

- In real situations there is very rarely only one form of energy transfer
  - Usually two or three happen at once

Tea cup example

- In the diagram below a more complex - and more 'real' - version of the situation above is shown
- Thermal energy is transferred **from hotter areas (the tea) to cooler areas (the cup, hands and air)** by the processes of:
  - **Conduction**; by **direct contact** between the tea and the **solid** sides of the cup and also by direct contact from the cup to the surface it is sitting on
  - **Convection**; from the **surface** of the coffee to the air **directly above** it
  - **Radiation**; from the sides of the **hot cup** in **all directions** to the surrounding air



In this example, heat is lost via conduction, convection and radiation

Wood fire example

- A wood (or coal) fire in a room heats it through **radiation** and **convection**
- As the fuel is so hot in a wood fire, it transfers a lot of thermal energy to the room through radiation
  - The fire transfers a much greater amount of thermal energy to **nearby** objects via radiation
- Air surrounding the fire is heated and rises, forming a convection current
  - This transfers thermal energy throughout the whole room

Car radiator example

- A car radiator transfers heat away from the engine, which reaches high temperatures
- A liquid travels between the radiator and the engine
  - When the liquid passes over the engine, it absorbs energy from the engine through **conduction**
  - This liquid then travels back to the radiator and transfers heat to the radiator, again by **conduction**
  - The radiator then transfers thermal energy to the surrounding air through **radiation**
- A car radiator is a dark colour, which allows it to emit more radiation
  - A large surface area also helps
- Once the radiator has absorbed thermal energy from the liquid, the liquid is cooler and the cycle begins again

Examiner Tips and Tricks

The specification requires you to have knowledge of energy transfers in:

- A wood or coal fire
- A car radiator

Familiarise yourself with how these systems work and their main forms of energy transfer.

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